Introduction

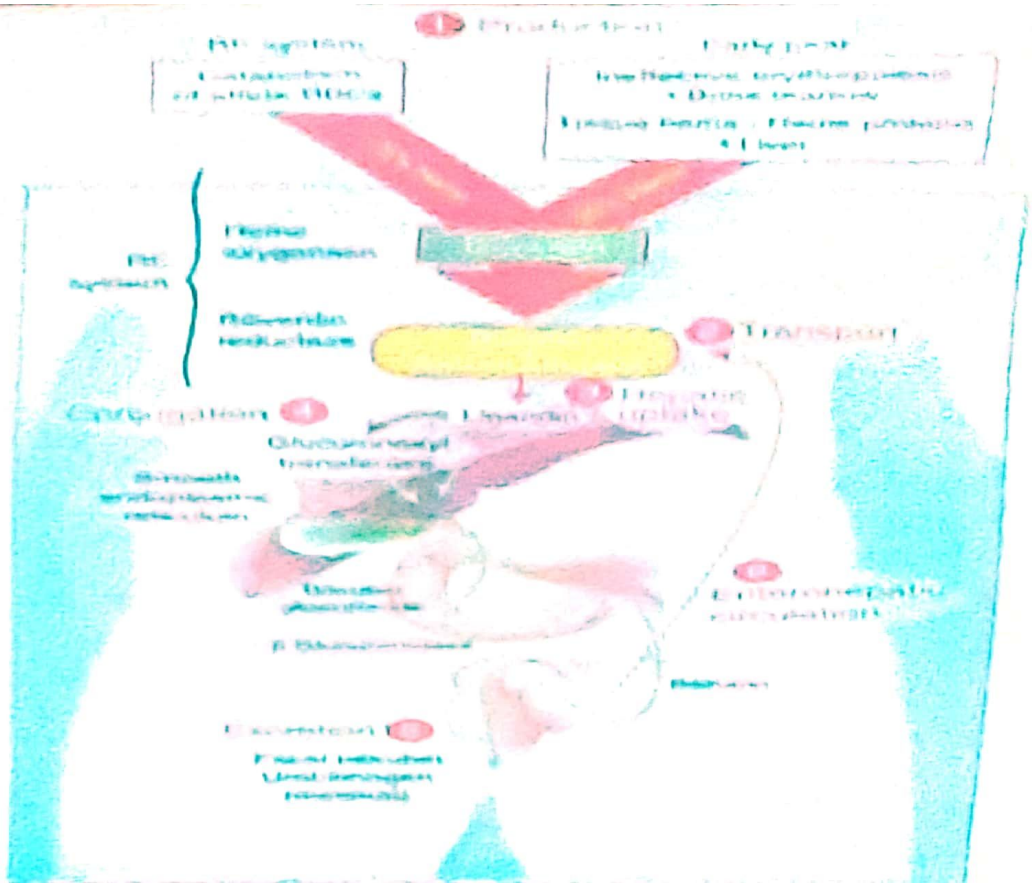
- Jaundice is a yellow discoloration of the skin and eyes caused by hyperbilirubinemia
- Jaundice usually becomes visible on the sclera at a level of 34 to 51 mcmol/L (2-3mg/dl) and on the face at about 68 to 86 mcmol/L.(4-5mg/dL)
- With increasing bilirubin levels, jaundice seems to advance in a head-to-foot direction, appearing at the umbilicus at about 15 mg/dL (258 mcmol/L) and at the feet at about 20 mg/dL (340 mcmol/L).

Scanned with CamScanner

Introduction

- Common (up to 60% term infants and 80% preterm infants)
  - Usually physiological- (important, however, to recognise the difference)
- Can potentially be toxic (mainly to the brain- kernicterus)
  - occurs mainly when unconjugated bilirubin crosses the BBB
  - Note that unconjugated jaundice may be physiological but conjugated jaundice (whilst not toxic) is almost always pathological

Scanned with CamScanner



Scanned with CamScanner

Neonatal Jaundice

Jaundice: is yellowish discoloration of skin and mucus membranes due to increased serum bilirubin above normal levels

* Normal cord bilirubin is less than 3 mg/dl.
* Jaundice is obvious clinically in neonate when serum bilirubin exceeds 5 mg/dl ; versus 3 mg /dl in adults.

Bilirubin Metabolism

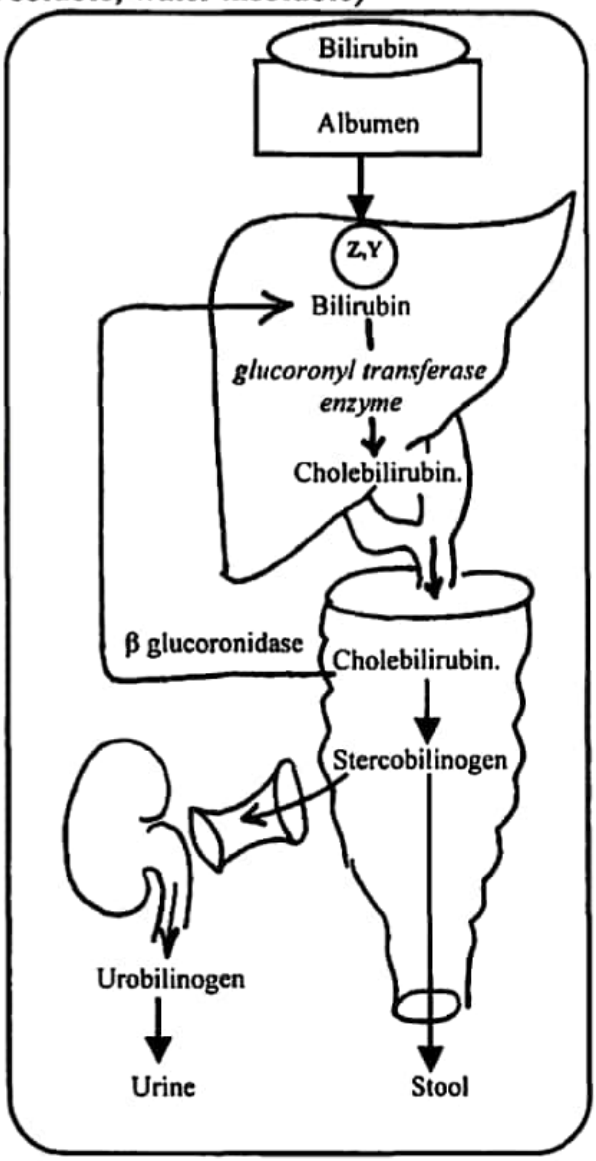
1- Production: Bilirubin is produced mainly from old RBCs

Old RBC's

Iron ← Heme ← | → Globin → Amino acid pool

Heme oxygenase ↓

Bilivirdin → Bilirubin (lipid soluble, water insoluble)



2- Transport:
Bilirubin is carried on albumin (so called unconjugated or hemebilirubin)

3- Uptake by hepatocytes:
Bilirubin bind to cytoplasmic ligandins ; Z & Y proteins to deliver it to endoplasmic reticulum where conjugation occur.

4- Conjugation:
Conjugation of bilirubin stimulated by glucoronyl transferase enzyme give rise to conjugated or cholebilirubin which is water soluble and lipid insoluble

5- Secretion:
Active secretion of conjugated bilirubin by liver cells into bile canaliculi.

vi- Excretion:
Excretion of conjugated bilirubin & bile salts into the intestine.

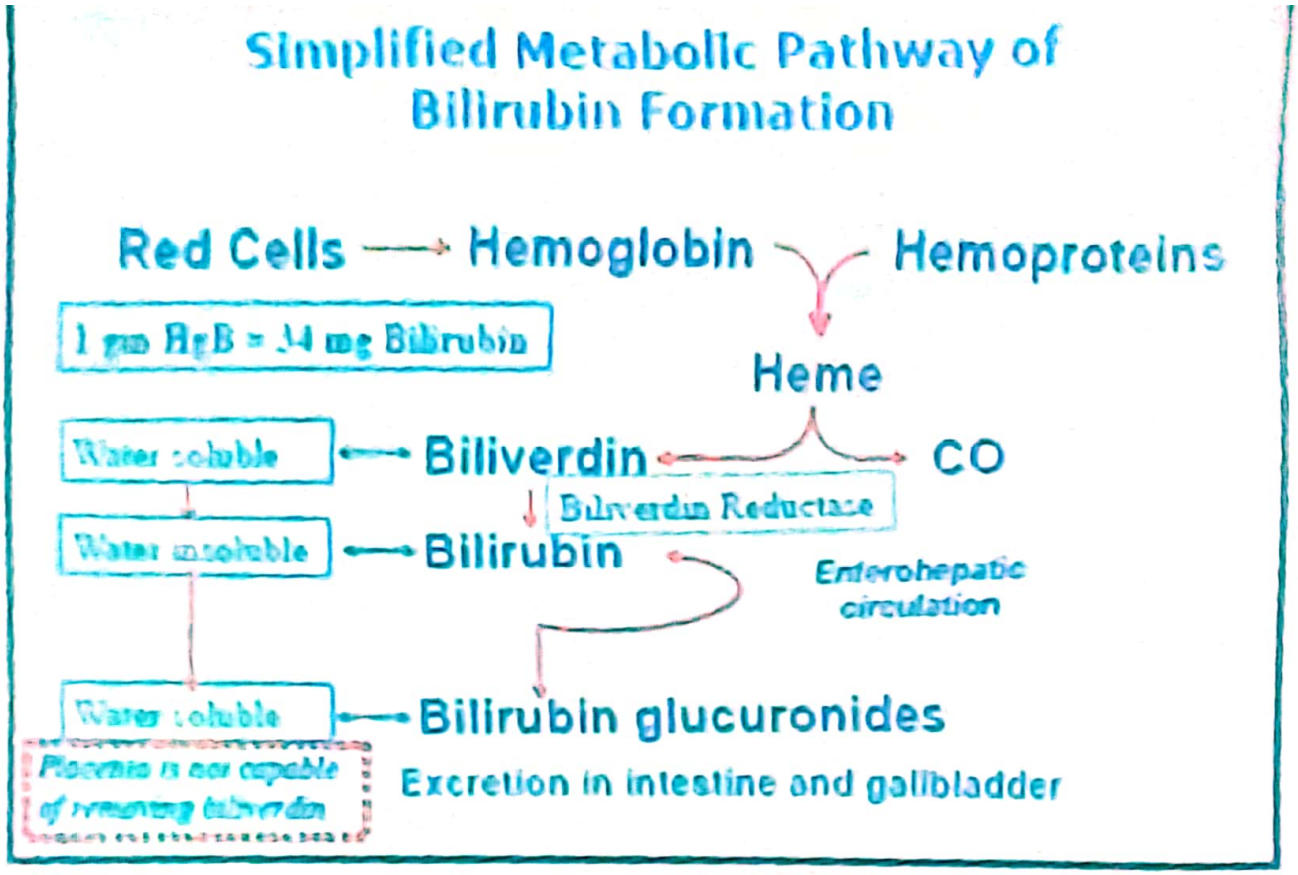
6- Bilirubin in intestine:

* Some amount is deconjugated by mucosal enzyme; β glucoronidase → unconjugated bilirubin → reabsorbed to the liver (entero- hepatic circulation)
* Some amount changed to stercobilinogen → stool
* Small amount of stercobilinogen reach the systemic blood (urobilinogen) → urine.

Scanned with CamScanner

- At birth, the placental connection is terminated, and although the neonatal liver continues to take up, conjugate, and excrete bilirubin into bile so it can be eliminated in the stool, neonates lack proper intestinal bacteria for oxidizing bilirubin to urobilinogen in the gut; consequently, unaltered bilirubin remains in the stool, imparting a typical bright-yellow color.
- Additionally, the neonatal GI tract (like that of the fetus) contains beta-glucuronidase, which deconjugates some of the bilirubin. Feedings invoke the gastrocolic reflex, and bilirubin is excreted in stool before most of it can be deconjugated and reabsorbed.
- However in many neonates, the unconjugated bilirubin is reabsorbed and returned to the circulation from the intestinal lumen (enterohepatic circulation of bilirubin), contributing to physiologic hyperbilirubinemia and jaundice

Scanned with CamScanner



Scanned with CamScanner

EVALUATION

History

- **History of present illness** should note age of onset and duration of jaundice. Important associated symptoms include lethargy and poor feeding (suggesting possible kernicterus), which may progress to stupor, hypotonia, or seizures and eventually to hypertonia. Patterns of feeding can be suggestive of possible breastfeeding failure or underfeeding. Therefore, history should include what the infant is being fed, how much and how frequently, urine and stool production (possible breastfeeding failure or underfeeding), how well the infant is latching on to the breast or taking the nipple of the bottle, whether the mother feels that her milk has come in, and whether the infant is swallowing during feedings and seems satiated after feedings.
- **Review of systems** should seek symptoms of causes, including respiratory distress, fever, and irritability or lethargy (sepsis); hypotonia and poor feeding (hypothyroidism, metabolic disorder); and repeated episodes of vomiting (intestinal obstruction).

Scanned with CamScanner

- **Past medical history** should focus on maternal infections (toxoplasmosis, other pathogens, rubella, cytomegalovirus, and herpes simplex [TORCH] infections), disorders that can cause early hyperbilirubinemia (maternal diabetes), maternal Rh factor and blood group (materno-fetal blood group incompatibility), and a history of a prolonged or difficult birth (hematoma or).
- Family history should note known inherited disorders that can cause jaundice, including G6PD deficiency, or other red cell enzyme deficiencies, thalassemias, and spherocytosis, and also any history of siblings who have had jaundice.
- Drug history should specifically note drugs that may promote jaundice (eg, ceftriaxone, sulfonamides [these drugs do not promote jaundice they instead lead to potentially more harm at lower measured bilirubin levels because they displace bilirubin from albumin increasing the free bilirubin fraction], and antimalarials

Scanned with CamScanner

Causes

Most cases involve unconjugated hyperbilirubinemia. Some of the most common causes of neonatal jaundice include

- Physiologic hyperbilirubinemia
- Breastfeeding jaundice
- Breast milk jaundice
- Pathologic hyperbilirubinemia due to hemolytic disease
- Liver dysfunction (eg, caused by parenteral alimentation causing cholestasis, neonatal sepsis, neonatal hepatitis) may cause a conjugated or mixed hyperbilirubinemia.

Scanned with CamScanner

Cholestasis

- Intrahepatic cholestasis
  - Sepsis
  - Hepatitis A
  - Hepatitis B
  - TORCH infections
  - Cystic fibrosis
- Extrahepatic cholestasis
  - Biliary atresia
  - Choledochal cyst
  - Tumors/strictures

Scanned with CamScanner

Intrahepatic pathologies

- Infectious hepatitis
- Metabolic diseases
  - Dubin-Johnson syndrome
  - Rotor syndrome
  - Galactosemia
  - Alpha-1-Antitrypsin deficiency
  - Hypermethioninemia
- Idiopathic neonatal hepatitis
- Alagille syndrome

Scanned with CamScanner

- **Risk factors**
- The following babies require closer post-natal observation for the development of jaundice
- Visible jaundice within 24 hours of birth
- Pre-term infants or low-birth weight babies
- A previous sibling who had neonatal jaundice requiring phototherapy
- A mother who intends to breast feed exclusively

Scanned with CamScanner

Physical examination

- Overall clinical appearance and vital signs are reviewed.
- The skin is inspected for extent of jaundice. Gentle pressure on the skin can help reveal the presence of jaundice. Also, ecchymoses or petechiae (suggestive of hemolytic anemia) are noted.
- The physical examination should focus on signs of causative disorders.
- The general appearance is inspected for plethora (maternofetal transfusion); macrosomia (maternal diabetes); lethargy or extreme irritability (sepsis or infection); and any dysmorphic features such as macroglossia (hypothyroidism) and flat nasal bridge or bilateral epicanthal folds (Down syndrome).
- For the head and neck examination, any bruising and swelling of the scalp consistent with a cephalohematoma are noted. Lungs are examined for crackles (rales), rhonchi, and decreased breath sounds (pneumonia). The abdomen is examined for distention, mass (hepatosplenomegaly), or pain (intestinal obstruction). Neurologic examination should focus on signs of hypotonia or weakness (metabolic disorder, hypothyroidism, sepsis).

Scanned with CamScanner